

POWDER Training Lecture Notes

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1 Introduction

The next few weeks of the POWDER training is meant to provide you with an understanding of the major tradeoffs and limitations caused by the wireless channel on communications and networking systems. With that understanding, you should know why the physical and multiple access layers are designed as they are currently, and be able to design the next generation of networking systems to achieve better reliability, fidelity, and robustness, and to provide new applications with their requirements.

The wireless channel is an analog channel. We use continuous-time, continuous-valued electromagnetic waves to repeatedly convey digital information. The channel strongly (and randomly) attenuates the signal, and adds in random thermal noise and interference from other transmitters. These real-world limitations introduce a few key challenges:

1. Matching: There is a mismatch between the analog medium and the digital information we are sending and receiving, and we must design methods to modulate (or translate) the digital information into an analog signal, and to demodulate (or translate back) the signal into digital information.
2. Loss: The channel introduces large losses into the signal that is sent. A receiver will amplify this signal, but will also amplify any noise and interference it receives. A designer must “budget” for this channel loss to build a reliable link.
3. Multipath: EM waves don’t travel only in a straight line. Antennas send signals to and received signals from all directions (although with different amplitudes as described by their antenna pattern). Waves interact with objects in the environment by changing direction, scattering into multiple directions, being attenuated by, and transmitting through them. These multiple signals arrive with different time delays, phases, and amplitudes.
4. Interference: Your link is only one of those using the radio channel at the same time, and even in the same frequency band. Systems should be designed to limit the interference to each other, and use the channel as efficiently as possible.

We will discuss these three key tradeoffs in the design of a wireless communication system: *efficiency*, *bandwidth*, and *fidelity*. Fidelity is the correctness of the received data (*i.e.*, the opposite of error rate). Bandwidth refers specifically to how much spectrum, in Hertz, is used by a communication network. Efficiency refers both to using as little power/energy, bandwidth, and time (latency) as possible. What is the trade-off between energy, bandwidth, and bit error rate? We want all three to be low (low consumption, low bandwidth usage, and low bit error rate). Energy and bandwidth are the costs of our communication system, and fidelity is how well it performs.

This section studies energy and power of our signals, from the transmitter to the receiver; including the loss and multipath limitations described above. On the topic of matching, today’s notes are *all about the analog medium*, that is, the continuous-time continuous-valued physical world of electromagnetic waves and voltage signals.

2 Power and Energy

Two of the biggest limitations in communications systems are (1) *energy / power*; and (2) *bandwidth*. This section provides some tools to deal with power and energy.

Recall that energy is power times time. **Use the units:** energy is measured in Joules (J); power is measured in Watts (W) which is the same as Joules/second (J/sec). Also, recall that our standard in signals and systems is define our signals, such as $x(t)$, as voltage signals (V). When we want to know the power of a signal we assume it is being dissipated in a 1 Ohm resistor, so $|x(t)|^2$ is the power dissipated at time t (since power is equal to the voltage squared divided by the resistance). The actual power in the voltage signal is scaled by the actual resistance of the load, but we ignore this constant for now.

A signal $x(t)$ has energy defined as

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

A signal with finite energy is called a *waveform* or *energy signal*. For some signals, E will be infinite because the signal is non-zero for an infinite duration of time (it is always on). These signals we call *power signals* and we compute their power as

$$P = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$$

2.1 Discrete-Time Signals

In the Rice book [6], it refers to discrete samples of the sampled signal x as $x(n)$. You may be more familiar with the $x[n]$ notation. We'll follow the Rice book's notation. Essentially, whenever you see a function of n (or k , l , m), it is a discrete-time function; whenever you see a function of t (or perhaps τ) it is a continuous-time function. I'm sorry this is not more obvious in the notation.

For discrete-time signals, energy and power are defined as:

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2 \tag{1}$$

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2 \tag{2}$$

3 Decibel Notation

We often use a decibel (dB) scale for power. If P_{lin} is the power in Watts, then the power in dBW is

$$P [\text{dBW}] = 10 \log_{10} \frac{P_{lin}}{1 \text{ W}}.$$

We convert from dB to linear by inverting the above formula,

$$P_{lin} = (1 \text{ W}) 10^{P[\text{dBW}]/10}.$$

Decibels are more general - they can apply to other unitless quantities as well, such as a gain (loss) $L(f)$ through a filter $H(f)$,

$$L(f) [\text{dB}] = 10 \log_{10} |H(f)|^2 \tag{3}$$

3.1 Power vs. Voltage Gain

Note that (3) could also be written as:

$$L(f) [\text{dB}] = 20 \log_{10} |H(f)| \quad (4)$$

Be careful with your use of 10 vs. 20 in the dB formula.

- Only use 20 as the multiplier if you are simultaneously converting linear to dB **and** from voltage to power; *i.e.*, taking the $\log_{10}$ of a voltage gain and expecting the result to be a dB power gain.

Our standard in communications systems design is to consider power gains and losses, not voltage gains and losses. So if we say, for example, the channel has a loss of 20 dB, this refers to a loss in power. In particular, the output of the channel has 100 times less power than the input to the channel.

Remember these three dB numbers:

- 3 dB: This means the number is double in linear terms.
- 10 dB: This means the number is ten times in linear terms.
- 1 dB: This means the number is a little over 25% more (multiply by 5/4) in linear terms.

With these three numbers, you can quickly convert losses or gains between linear and dB units without a calculator. Just convert any dB number into a sum of multiples of 10, 3, and 1.

Example: Convert dB to linear values:

1. 30 dBW
2. 33 dBm
3. -20 dB
4. 4 dB

Solution:

1. $(1 \text{ W})10^{30[\text{dBW}]/10} = 10^3 \text{ W} = 1000 \text{ W}.$
2. $33 \text{ dBm} = 30 \text{ dBm} + 3 \text{ dB} = 1000 \text{ mW} \times 2 = 2 \text{ W}.$
3. $-20 \text{ dB} = 10^{-20[\text{dBW}]/10} = 10^{-2} = 0.01.$
4. $4 \text{ dB} = 3 \text{ dB} + 1 \text{ dB} \approx 2(1.25) = 2.5.$

Example: Convert linear values to dB:

1. 0.2 W

2. 40 mW

Solution:

1. $0.2 \text{ W} = (0.1\text{W})(2) = -10 [\text{dBW}] + 3 [\text{dB}] = -7 \text{ dBW}$
2. $40 \text{ mW} = (10\text{mW})(2)(2) = 10 [\text{dBm}] + 3 [\text{dB}] + 3 [\text{dB}] = 16 [\text{dBm}]$.

Example: Convert power relationships to dB:

Convert the expression to one which involves only dB terms.

1. $P_{y,lin} = 100P_{x,lin}$
2. $P_{o,lin} = G_{connector,lin} L_{cable,lin}^{-d}$, where $P_{o,lin}$ is the received power in a fiber-optic link, where d is the cable length (typically in units of km), $G_{connector,lin}$ is the gain in any connectors, and $L_{cable,lin}$ is a loss in a 1 km cable.
3. $P_{r,lin} = P_{t,lin} \frac{G_{t,lin} G_{r,lin} \lambda^2}{(4\pi d)^2}$, where λ is the wavelength (m), d is the path length (m), and $G_{t,lin}$ and $G_{r,lin}$ are the linear gains in the antennas, $P_{t,lin}$ is the transmit power (W) and $P_{r,lin}$ is the received power (W). This is the Friis free space path loss formula.

These last two are what we will need in Rice Section 6.4 [6], when we discuss link budgets. The main idea is that we have a limited amount of power which will be available at the receiver.

4 Antennas

Antennas are passive – they don’t create more RF power, in total. They’re also not isotropic – don’t send power equally in every direction. Every wireless link between a transmitter and receiver will experience two *antenna gains* that are a function of the direction of departure (at the transmitter antenna) and the direction of arrival (at the receiver antenna). Some angles get a high(er) gain, and some angles get a lower gain.

We define the power gain G in direction (θ, ϕ) as $G(\theta, \phi)$. Here $G(\theta, \phi)$ is a linear power gain that will multiply the signal power. Typically antenna gain patterns are given by spec sheets in dB. Antenna gains are unitless. Nevertheless, the unit is reported as “dBi” where the “i” stands for isotropic, the non-existent antenna that has equal gain of 1 in all directions. I do not use the “i” in my notation.

Knowing the angular direction between the transmitter and receiver in elevation and in the horizontal plane, allows us to use $G(\theta, \phi)$ to calculate how much the signal power will experience gain along its link. A graphical example is shown in Figure We use $G_T(\theta, \phi)$ to denote the transmitter antenna gain pattern, and $G_R(\theta, \phi)$ for the receiver. They often use different kinds of antennas, so these can be very different patterns.

Antenna patterns are all directional to some extent, since they are not isotropic. Antennas tend to be summarized by

- the maximum of $G_T(\theta, \phi)$, which is confusingly called the “gain”;
- the vertical and horizontal beamwidths. The horizontal beamwidth is the width of angles θ that are more than -3 dB down from the maximum gain; the vertical beamwidth is the width of angles ϕ that are more than -3 dB down from the maximum gain

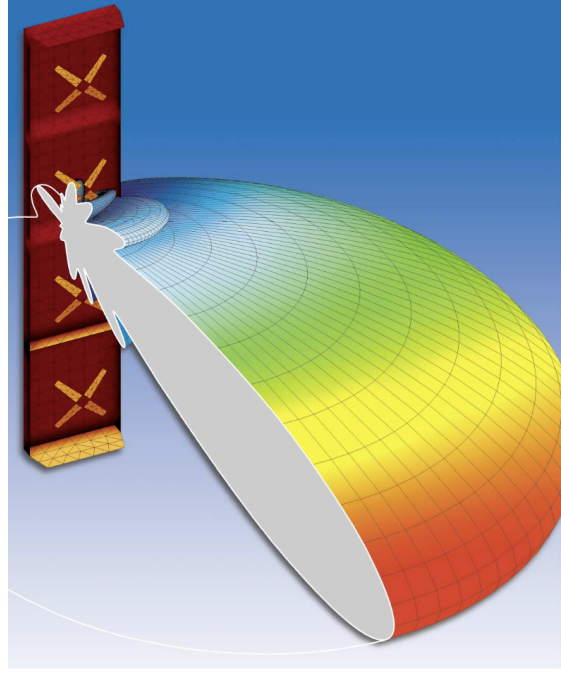


Figure 1: Image from cover of 2001 *Katherin Antennas* catalog of (half of) an gain pattern $G_T(\theta, \phi)$ for a cellular sector antenna. The cut shows the vertical gain pattern in the mainlobe.

Typically, the gain is a function of dimension of the antenna on the same axis, and inversely prop to the wavelength, although this is not always the case. Tall antennas have a narrow horizontal beam. When we use low frequencies, this means we have a large wavelength, and thus we need a large antenna to achieve the same gain.

5 Mean Received Power Models

Given that we want our wireless communication system to operate in some application or for some link, how can we calculate how much power will be received, typically? We do this here for wireless (and ignore wired) channels. We divide wireless channels into free space channels, and obstructed (non-free-space channels).

5.1 Free Space

‘Free space’ is the idealization in which nothing exists anywhere near the line between the transmitter and receiver, and can really only be used for deep space and satellite communications (although satellite communications systems will need to separately account for weather losses in the atmosphere.) In addition, this formula serves as a starting point for other radio propagation formulas. Note the received power is called *both* P_R and C , equivalently. In free space the received power is calculated from the Friis formula,

$$C = P_R = P_T G_T G_R \left(\frac{\lambda}{4\pi R_0} \right)^2 \left(\frac{R_0}{R} \right)^2 \quad (5)$$

where

- G_T and G_R are the antenna gains at the transmitter and receiver, respectively. These are unitless. The gains are actually in terms of the angle of departure and angle of arrival at the transmitter and receiver, but to keep the notation simple, I leave those out.
- P_T is the transmit power, in Watts.
- λ is the wavelength at signal frequency. For any signal that is not called “ultrawideband”, the wavelength is nearly constant across the bandwidth, so we just use the center frequency f_c . Note that $\lambda = c/f_c$ where $c = 3 \times 10^8$ meters per second is the speed of light.
- R_0 is a reference distance, say 1 meter or 10 meters. You may notice that the R_0^2 variables cancel, so why have I put them in? It is useful for understanding the relationships and for keeping track of the units. 1) One can measure the received power at a reference distance R_0 during testing, and then replace the $P_T G_T G_R \left(\frac{\lambda}{4\pi R_0}\right)^2$ part with the measurement; 2) Each fraction has units that cancel; and 3) We consider the $\left(\frac{R_0}{R}\right)^2$ part as the *path loss*, i.e., loss due to the actual path length (distance between the transmitter and receiver).
- Note that the formula is named after a Danish-American engineer Harald T. Friis [2], which just coincidentally sounds like “Free”, but you may hear some confusion about what the formula is called.

In (5), everything is in linear terms. Typically communication systems engineers use decibels to express these numbers. As a reminder and we write P_R [dBm] or G_T [dB], and their values are given by:

$$\begin{aligned} P_R [\text{dBm}] &= 10 \log_{10} \frac{P_R}{1 \text{ mW}} \\ G_T [\text{dB}] &= 10 \log_{10} G_T \end{aligned}$$

The Friis formula of (5), given in dB, is

$$C [\text{dBm}] = G_T [\text{dB}] + G_R [\text{dB}] + P_T [\text{dBm}] + 20 \log_{10} \frac{\lambda}{4\pi R_0} - 20 \log_{10} \frac{R}{R_0} \quad (6)$$

This says the dB received power, C [dBm], is linearly proportional to the log of the distance R .

The dB received power at the reference distance R_0 is called P_0 [dBm] and is given by:

$$P_0 [\text{dBm}] = G_T [\text{dB}] + G_R [\text{dB}] + P_T [\text{dBm}] + 20 \log_{10} \frac{\lambda}{4\pi R_0} \quad (7)$$

Again, P_0 [dBm] is something you might measure with a well-calibrated receiver and an antenna. If you know (or estimate) P_0 [dBm], then the relationship of received power (dBm) with distance R is simply:

$$C [\text{dBm}] = P_0 [\text{dBm}] - 20 \log_{10} \frac{R}{R_0}. \quad (8)$$

There are also typically other losses in a transmitter / receiver system; losses in a cable, other imperfections, etc. In the Rice book, these are labelled as L [dB] [6]. You must subtract the dB loss from the left-hand side of (6):

$$C [\text{dBm}] = P_0 [\text{dBm}] - 20 \log_{10} \frac{R}{R_0} - L [\text{dB}]. \quad (9)$$

Note that you need to be careful (even [6] is sometimes ambiguous) that if you call it a dB “loss”, it is value that you subtract from (8). Some people will call it a gain $G[\text{dB}] = -L[\text{dB}]$, but you if you call it a “gain” than add $G[\text{dB}]$ to (8).

5.2 Non-free-space Channels

We don’t study a large variety of radio propagation path loss formulas in this course. But, a very short summary is that radio propagation on Earth is different than the Friis formula suggests. Many other formulas exist that approximate the received power as a function of distance, antenna heights, type of environment, etc. The main problem these approximations address is that walls, trees, buildings, and varying terrain decrease the power that would be received as compared to the Friis model. This is called “shadowing” in analogy to how objects shadow light.

The most common model for real world shadowing effects is the path loss exponent model, in which received power in dBm is linear with the log of distance,

$$C[\text{dBm}] = P_0[\text{dBm}] - 10n_p \log_{10} \frac{R}{R_0}, \quad (10)$$

for some constant n_p . Here, $P_0[\text{dBm}]$ is typically the same as we described earlier, that is, the path loss that the Friis model would predict at some short distance R_0 from the antenna. Usually R_0 is a very short distance, like 1 m or 10 m. Effectively, because of shadowing caused by buildings, trees, etc., the average loss may increase more quickly than $1/R^2$, instead, it may be more like $1/R^{n_p}$.

Equivalently, in linear terms:

$$C = P_R = P_T G_T G_R \left(\frac{\lambda}{4\pi R_0} \right)^2 \left(\frac{R_0}{R} \right)^{n_p}$$

This is a model that comes with lots of evidence from empirical measurement studies, for example, a study I did shown in Figure 2. In dense urban areas people observe a much higher n_p than in rural areas because of losses due to buildings. In mountainous areas, people observe a higher n_p than in flat areas because of the losses due to the signal diffracting over or around mountains. When the antennas are both close to the ground, the n_p is higher than when the antennas are held tens of meters away from the ground, due to ground reflection effects. Finally, in buildings, we observe a higher n_p when the walls are made of bricks or concrete, than when the walls are drywall and wood frame. The value of $n_p = 2.3$ in Figure 2 is likely so low because the environment had only soft-partitioned cubicle walls, and no solid walls. The value of n_p may range from 2 to 4.5, with some values going lower than 2 or higher than 4.5. In real life, you would want to measure n_p in the type of environment you want to have your link operate in, or use values someone else has measured from the literature on measurement-based path loss models, e.g. Hashemi’s review of indoor channel models [4].

However, note that (10) is only the *average* received power. In reality, any one particular measurement of received power a distance R from the transmitter may be off by 0-30 dB from the average, due to differences in the particular path that the signal travels, due to small-scale fading, and for other reasons.

6 Multipath Fading

If I had to name the most important principle I know about wireless, I’d have to say it is mechanism behind the delay and frequency dependence of the multipath channel gain.

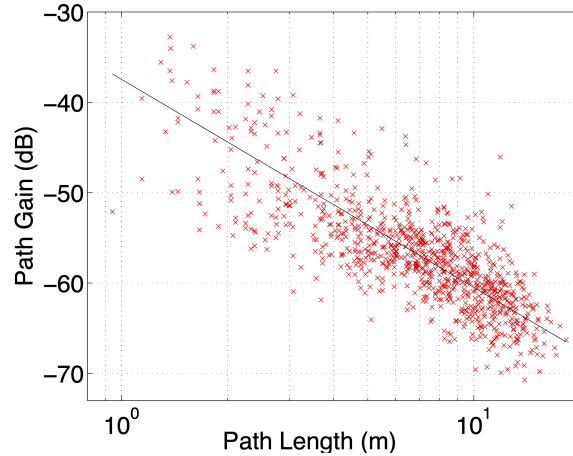


Figure 2: Measured ‘path gain’, $C [\text{dBm}] - P_T [\text{dBm}]$, for an indoor office area in the 2.4 GHz band, averaged over a 40 MHz band. The line fit is for $n_p = 2.3$. The standard deviation of error around the regression line fit is 3.9 dB [5].

Multipath is the phenomenon that multiple electromagnetic waves, each arriving from different directions, arrives at the receiver antenna. Each wave induces a current signal, or equivalently, a voltage signal, on the antenna because it is a conductor. These voltage signals add together.

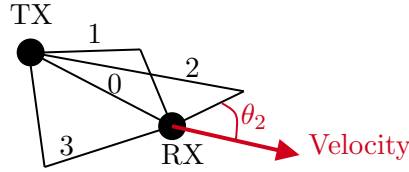


Figure 3: Example channel with $L = 4$ multipath components between the transmitter and receiver. The receiver is moving at an angle of θ_l with respect to multipath component l .

Let’s first consider that the transmitted signal is a constant-amplitude cosine at frequency f . This is what is called a continuous wave (CW) signal. The received voltage signal is a sum of time-delayed, attenuated sinusoids at that frequency f . Recall that we represent the received signal in complex baseband notation. To ease our way into this notation, instead of writing the signal as a cosine at frequency f and power V^2 , we write it as

$$s(t) = \sqrt{2}\Re \left\{ V e^{j2\pi f t} \right\}. \quad (11)$$

One can see that $s(t) = \sqrt{2}V \cos(2\pi f t)$. The complex baseband signal is everything multiplying the $e^{j2\pi f t}$ inside the real operator. That is, the transmitted complex baseband signal is simply V .

But there are multiple paths which the EM wave travels to the receiver, each with a different time delay. Let’s say there are L multipath components, numbered 0 through $L - 1$. Component l has amplitude gain α_l and time delay of τ_l . Consider the received signal for the l th multipath is changed both by the amplitude gain and the time delay:

$$v(t) = \sqrt{2}\Re \left\{ V \alpha_l e^{j2\pi f (t - \tau_l)} \right\} \quad (12)$$

$$= \sqrt{2}\Re \left\{ V \alpha_l e^{-j2\pi f \tau_l} e^{j2\pi f t} \right\}. \quad (13)$$

The complex baseband version of this received signal for the l th path is thus

$$v_l^{CB}(t) = V\alpha_l e^{-j2\pi f\tau_l} \quad (14)$$

Then the total complex baseband voltage at the receiver antenna, which I'm calling V_{TOT} , is:

$$V_{TOT} = \sum_{l=0}^{L-1} V\alpha_l e^{-j2\pi f\tau_l} \quad (15)$$

If you're familiar with the Fourier transform, you may remember the property that a time delay of τ_l in the time domain corresponds to a phase change of

$$\phi_l = -2\pi f\tau_l. \quad (16)$$

This is the same thing happening in (16).

Each multipath travels a different distance, thus its τ_l is different. Plus, multipath component l arrives from a different angle with respect to the receive antenna or the direction of motion of the receiver (as depicted in Figure 3).

The phases $\{\phi_l\}_l$ of these multipath components change as the antenna on a mobile device takes different positions. The delays $\{\tau_l\}_l$ are changing at different rates while the receiver moves. To be more precise, the delay τ_l changes over time proportional to the receiver's speed relative to the transmitter, which means it is proportional to the cosine of the angle θ_l between its angle of arrival and the angle at which the receiver is moving, as drawn in Figure 3.

- See video <https://www.youtube.com/watch?v=5DTwxr526Vg> at time 3:06

For example in Figure 3 the propagation delay of path 0, τ_0 , is increasing at a faster rate than the propagation delay of path 1, τ_1 , because $|\cos(\theta_0)| > |\cos(\theta_1)|$. Because θ_l is different for each multipath, the complex value $e^{j\phi_l}$ rotates around the origin at a different rate (and perhaps in a different direction). The impact on the sum in (15) is that the total voltage is this time-varying sum of complex-valued multipath components, each rotating in phase with a different rate.

- See video <https://www.youtube.com/watch?v=5DTwxr526Vg> at time 3:40

Please see the lecture video again to watch an example of this sum over time for $L = 8$ equal amplitude paths at time 4:19. The sum appears random, with a changing magnitude (distance from the origin) and power (squared distance from the origin).

If you've seen a "Spirograph" (the kids art toy), you're familiar with the sum of two vectors with different amplitudes α_l adding with phases changing at different rates. If not, a photo from an online 3D printed set of spirograph gears in Figure 4 gives you an idea of what the toy produces – rotating the gears while a pen is held by one gear at a particular relative position with respect to the center of that gear means that the pen draws out the sum of two rotating vectors, each rotating at a different rate. Since the gear ratio is a fraction (of two integers), it repeats after some number of cycles.

The point about this to notice is that the power of the received signal varies as a function of the position of the antenna. Sometimes, the complex amplitudes have similar phases and add constructively. Other times, these complex amplitudes are nearly opposite in phase, and when added, are at or near the origin. The effect of multipath fading on the received power ($|V_{TOT}|^2$) is that it varies from a few dB gain to a 30 dB loss as the antenna moves on the order of a quarter of a wavelength.

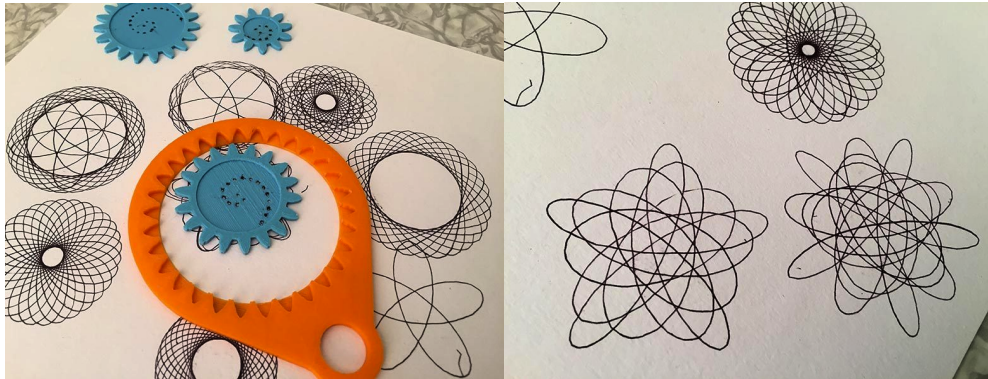


Figure 4: These spirographs, from 3D printed gears [1] allow you to draw the sum (or difference) of two or more vectors, each rotating at different rates. While any short segment seems random, viewing the entire path shows the pattern in the sum over time.

Note the difference between the antenna “moving” and the antenna “taking different positions”. The change in phase comes from the antenna being at different positions, not because it has a non-zero velocity, by itself. Obviously having a non-zero velocity moves the antenna to different positions over time. But the same differences would be observed by multiple stationary antennas at different positions. Using multiple antennas for one transceiver is a key method to deal with fading. For a great explanation of using multiple antennas at the transmitter and receiver (MIMO) written for networking area researchers, see [3].

6.1 Distribution Models

With multipath fading so random, and so significant, how can we design a reliable system? This section describes what we can quantify about multipath fading that then allows us to design reliable wireless links.

To proceed with this quantification, we need probabilistic analysis, which requires us to move from considering purely “specular” multipath to combinations of specular and “diffuse” multipath:

- *Specular multipath*: What we’ve been talking about: individual multipath components, of which we said there were M total, each with its own amplitude and phase.
- *Diffuse multipath*: multipath which are each infinitely low in power, but there are infinitely many of them. Typically used to simulate the many many multipath that do arrive due to scattering and diffraction. It is easier to talk about fading when we can lump lots and lots of multipath into this “diffuse” camp, rather than specifying each individual multipath and how much power each one has.

6.2 Rayleigh Fading

Main idea: all multipath are diffuse. These diffuse multipath have small amplitude and random phase. Using the central limit theorem, we can come up with a pdf of the magnitude of the received voltage, $r = |V_{TOT}|$. In this case, we can derive the random variable R for the received voltage, and it is called the Rayleigh distribution, named after Lord Rayleigh, aka John William Strutt, a British physicist. This is why the fading is called “Rayleigh fading”. We use Rayleigh fading as a model when there are lots of multipath with approximately equal power.

6.3 Rician fading

Main idea: One specular path with amplitude $V_0 = A$ plus diffuse power.

The Rice distribution is named after its creator, Stephen O. Rice. Note that people just can't agree on the spelling of the adjective made out of his name, whether it is "Rician" or "Ricean".

The Rician K -factor is defined as the ratio between the power in the specular component to the power in the diffuse components,

$$K = \frac{A^2}{2\sigma^2}, \quad \text{or } K(\text{dB}) = 10 \log_{10} \frac{A^2}{2\sigma^2}. \quad (17)$$

Note that for $K = 0$ or $K(\text{dB}) = -\infty$, the Rician pdf is the same as the Rayleigh pdf.

6.4 Fade Margin

Def'n: *Fade Margin*

the additional power loss which is included in the link budget in order to keep the received signal power higher than the minimum, within a given probability.

There is a chance, no matter what fade margin is chosen, that the actual received power will be even lower than that margin below the average received power. I include in Figure 5 a figure with multiple values of K for Rician fading. It also includes Rayleigh, because Rayleigh is the same as Rician when $K = -\infty$.

For example, for Rayleigh fading, one could set a fade margin of 10 dB, but there is a 7% chance that the signal experiences a fade worse than 10 dB; one could set a 28 dB fade margin, and there is a 0.1% chance that the signal experiences a fade worse than 28 dB in a Rayleigh fading multipath channel. This is a very severe problem for mobile communications since link budgets are tight (as you have seen) and there is not generally 30 dB to spare to make the link reliable even when fading is at its worst.

Example: Fade margin design:

Assume that it is acceptable to have the received power fall below the receiver sensitivity 1% of the time, and that the channel is Rayleigh. What fade margin (in dB) is required? If that were a Rician fading channel with $K = 6$ dB, what fade margin would be required?

Solution: We are given $P[R \leq r] = 0.01$. So we look at the 1% line on the y axis. For the Rayleigh case ($K = -\infty$), I read about 18.5 dB. For the Rician fading channel with $K = 6$ dB, I read about 11 dB.

Example: Fade margin design: Rician case

Assume that it is acceptable to have the received power fall below the receiver sensitivity 1% of the time, and that the channel is Rician, with K -factor 3 dB, 9 dB, or 15 dB. What fade margin (in dB) is required?

Solution: From the chart, on the horizontal line from 1%, the fade margin would be -15.5 dB, -7.0 dB, and -3 dB, for K -factor of 3 dB, 9 dB, and 15 dB, respectively.

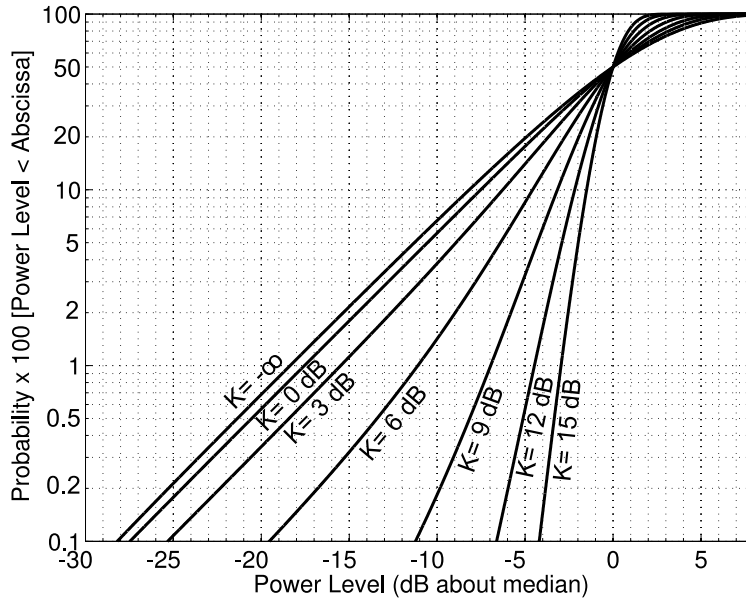


Figure 5: Rician CDF for various K -factors. Note “Power Level” is $20 \log_{10} \frac{r}{r_{median}}$, and the y-axis is the probability $\times 100$ (percentage) that the power level (fading gain compared to the median) is less than the value on the abscissa.

6.5 Frequency Selectivity

Note also that if the frequency f in (16) changes, it has a similar impact to a moving antenna. That is, it causes different multipath l to change phase at different rates, for the same change in frequency, because τ_l are different for each l . Thus multipath fading is also called *frequency selective*.

In particular, we can simply observe that (15) is a function of the frequency f . If we were to send different frequencies, the received voltage would be different in phase and amplitude:

$$V_{TOT}(f) = \sum_{l=0}^{L-1} V \alpha_l e^{-j2\pi f \tau_l} \quad (18)$$

We call the channel gain as a function of frequency, i.e., $V_{TOT}(f)/V$, as the channel frequency response (CFR):

$$H(f) = V_{TOT}(f)/V = \sum_{l=0}^{L-1} \alpha_l e^{-j2\pi f \tau_l} \quad (19)$$

7 Signal/Noise Energy and Power Ratios

The probability of bit error for a digital modulation is typically written as a function of $\frac{\mathcal{E}_b}{N_0}$. Here, the noise energy is N_0 , and the bit energy is $\mathcal{E}_b$. This is the most critical energy ratio that we need in the design of digital communications systems because the probability of bit error is always a function of $\frac{\mathcal{E}_b}{N_0}$.

7.1 Noise Energy

The noise energy N_0 that adds to the signal at the receiver can be calculated as:

$$N_0 = kT_{eq}$$

where $k = 1.38 \times 10^{-23}$ J/K is Boltzmann's constant and T_{eq} is called the equivalent noise temperature in Kelvin. How do you find the equivalent noise temperature for a receiver? Answering this question is a topic covered in another course, and you are not responsible for that material here. But in short, the equivalent temperature is a function of the receiver circuit design, and T_0 , the temperature of the environment at which the antenna is pointing at. This temperature T_{eq} is always going to be higher than T_0 . Basically, all receiver circuits add noise to the received signal. With proper design (and more expensive and power-hungry components), T_{eq} can be kept low.

7.2 Signal Energy

To me, this is a critical thing to understand. Given a constant noise energy, there is a certain amount of energy, in Joules, that is needed to convey each bit. There's no real way around this – as data rates increase, we will necessarily require more energy because we're sending more bits.

However, most engineers are more familiar with a power ratio, the signal to noise power ratio, or SNR. This next section talks about the relationship between SNR, bit rate, bandwidth, and $\frac{\mathcal{E}_b}{N_0}$.

We can write $\mathcal{E}_b = CT_b$, since energy is power $\times$ time. To separate the effect of T_b , we often denote:

$$\frac{\mathcal{E}_b}{N_0} = \frac{C}{N_0} T_b = \frac{C/N_0}{R_b} \quad (20)$$

where $R_b = 1/T_b$ is the bit rate. In other words, $C/N_0 = \frac{\mathcal{E}_b}{N_0} R_b$. This is signal power to noise energy ratio.

Next, let's convert into a signal power to noise power ratio. The noise power $N = N_0 B$, where B is the bandwidth of the receiver. This bandwidth is affected by how wide of a frequency band the antenna and receiver filters allow in, not directly about the *signal's* bandwidth. However, we typically assume that the receiver is designed to allow in frequencies only within the desired signal's band. Plugging in,

$$\frac{\mathcal{E}_b}{N_0} = \frac{C}{N/B} \frac{1}{R_b} = \frac{C}{N} \cdot \frac{B}{R_b} \quad (21)$$

Note $\frac{C}{N}$ is the signal power to noise power ratio, in short, the SNR.

So our key relationship is that:

$$\frac{\mathcal{E}_b}{N_0} = \frac{C}{N} \cdot \frac{B}{R_b}. \quad (22)$$

We'll be measuring the C/N for our POWDER links in our measurement lab. You'll see that that C/N imposes a limit on the possible values of $\frac{\mathcal{E}_b}{N_0}$. Later, we'll connect the $\frac{\mathcal{E}_b}{N_0}$ to receiver probability of bit error.

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